Cote 140-3

and likewise † [m1] [scan 492]

Marginal note. [m1] $\begin{cases} \Psi^{\pi}_{-\bar{\mathrm{j}}|-\mathrm{j}} : \widetilde{\Gamma}'_{\mathbb{Q}} \simeq \widetilde{\mathcal{T}}'_{0,3} \longrightarrow T_{\rho}(\widehat{\mathbb{Z}})/\pm 1 \\ \widetilde{\Gamma}'_{\mathbb{Q}} = \{\gamma \in \widetilde{\Gamma}_{\mathbb{Q}} \mid \varepsilon_3(\mu) = 1, \text{ where } \mu = \chi(\gamma)\} \end{cases}$

We thus see that the indeterminacies of the signs for $\Theta^{\pi}_{\rho}, \Theta^{\pi}_{\sigma}$ correspond precisely to the indeterminacies for Θ_G itself — everything hangs together! The latter is removed † tautologically when we replace $\widetilde{M}_{0,3}$ by the † ad-hoc extension $\widetilde{\mathcal{T}}_{1,1}$ by ± 1, so that we find [m3] (50

Marginal note. [m3] We will note however that Θ^{π}_{ρ}, unlike the splitting $\Psi^{\pi}_{\rho} = \Psi^{\pi}_{1/2}$, is defined on $\widetilde{\mathcal{T}}_{0,3}$ in its entirety.

$$(50) \qquad \widetilde{\mathcal{T}}_{1,1} \longrightarrow \mathrm{GL}(2, \widehat{\mathbb{Z}})$$

and one should † interpret this homomorphism as the fundamental "modular" homomorphism of the theory of elliptic curves (that the inverse image of † $\Gamma_{\mathbb{Q}} \subset \tilde{\Gamma}_{\mathbb{Q}}$ in $\widetilde{\mathcal{T}}_{1,1}$ can be interpreted as $\mathcal{T}_{1,1} \simeq \pi_1(M_{1,1\,\mathbb{Q}})$ must be more or less tautological, by construction.)

It is clear in any case that the two coincide for every $\gamma \in \widetilde{M}_{0,3}$ viewed as|when elements of

$$(51) \qquad \mu_2(\widehat{\mathbb{Z}}) = \{\pm 1\}^{\mathbb{P}} \qquad (\mathbb{P} = \text{set of prime numbers})$$

[p. 491]

Uncertainty retained from cancelled passages.

140-3-s492-D001 $\Psi^{\pi}_{-\mathrm{j}}$
140-3-s492-D002 by [illeg. 2]
140-3-s492-D003 covering|profinite with|of 2 sheets
140-3-s492-D004 be able to
140-3-s492-D005 Γ

i.e. they coincide modulo a homomorphism [scan 493] [p. 549]

$$\dagger\tilde{\Gamma}_{\mathbb{Q}} \longrightarrow \mu_2(\widehat{\mathbb{Z}}) \simeq \{\pm 1\}^{\mathbb{P}}$$

i.e. a homomorphism

$$G_{\mathrm{m}}(\widehat{\mathbb{Z}})_2 = (\widehat{\mathbb{Z}}^*)_2 \simeq \{\pm 1\}^{\mathbb{P}} \longrightarrow \dagger_2 G_{\mathrm{m}}(\widehat{\mathbb{Z}}) = \mu_2(\widehat{\mathbb{Z}}) \simeq \{\pm 1\}^{\mathbb{P}}$$

— so it is not a certain indeterminacy that is at issue! It is morally certain that the two coincide — I have not tried to prove it.

As for (35) (36), they must surely correspond to the action of $\Gamma_{\mathbb{Q}}$ on the Tate modules of elliptic curves with exceptional isomorphisms, corresponding precisely to the points $(1/2, 2, -1)$ and $(-\bar{\mathrm{j}}, -\mathrm{j})$ of $M_{0,3\,\mathbb{Q}} = M_{1,1}[2]'_{\mathbb{Q}}$. The indeterminacy of sign must|of which correspond|corresponds to the indeterminacy in lifting a point of $M_{1,1}[2]'_{\mathbb{Q}}$ to a point of $M_{1,1}[2]_{\mathbb{Q}}$, i.e. in choosing an elliptic curve (with "rigidification" of level 2) corresponding to the Legendre data $(0, 1, \infty, q|\rho)$ with $\rho|q = 1/2$ or $\rho|q = -\bar{\mathrm{j}}$. [p. 492]

Uncertainty retained from cancelled passages.

140-3-s493-D001 $M_{0,3}$

140-3-s493-D002 μ

Even once the ambiguity of sign for Θ^{π}_{G} is removed by passing to the two-sheeted extension $\widetilde{\mathcal{T}}_{1,1}$ of $\widetilde{M}_{0,3}$, the indeterminacy of the signs for $\Theta^{\pi}_{\rho}, \Theta^{\pi}_{\sigma}$ is not removed for all that, for the splittings † $\Psi^{\pi}_{1/2}, \Psi^{\pi}_{-\bar{\jmath}}$) are not for all that splittings (even partial!) of † the extension $\widetilde{\mathcal{T}}_{1,1}$, but only of $\widetilde{M}_{0,3}$ — and lifting them to splittings of $\widetilde{\mathcal{T}}_{1,1}$ over $\mathcal{T}_{1,1}$, amounts, very precisely, to choosing precisely elliptic curves with Legendre rigidification over the points $1/2, -\bar{\jmath}$. Thus, calculations made in terms of automorphisms of the groups with loops $\widehat{\Pi}_{0,3}$ or of $\mathcal{C}^{+\wedge}_{0,3}$, give back (without having looked for them!) results that fit very closely with particularly crucial arithmetico-geometric truths. [scan 494] [p. 493]

Uncertainty retained from cancelled passages.

140-3-s494-D001 $\Psi^{\pi}_{\rho}, \Psi^{\pi}_{\sigma}$

140-3-s494-D002 this

49. Homomorphisms of $M_{0,3}$, the general groups $M_{\rho,\sigma}$. [scan 495] [p. 550]

m5

Where we get our feet wet…

Marginal note. m5 All of this section I seems entirely useless for what follows, it is a preliminary groping to lead into section II page 553.

I) Consider a closed subgroup G of $M_{0,3}$, containing $\mathcal{C}^{+\wedge}_{0,3}$, hence the inverse image of a closed subgroup of $\tilde{\Gamma}_{0,3}$. We assume that the latter is "big enough" — say that it contains a subgroup of finite index in $\Gamma_{\mathbb{Q}} \subset \tilde{\Gamma}_{0,3}$. For the moment, we assume for simplicity that $G \subset \operatorname{Ker} \varepsilon_3$. In summary

$$(1) \qquad \mathcal{C}^{+\wedge}_{0,3} \subset G \subset \widetilde{M}_{0,3}, \qquad u \in G \Longrightarrow \mu(u) \equiv 1 \ (3).$$

This last hypothesis thus means that for every $u \in G$, we have $\alpha_0(u) \in \mathcal{C}^{\wedge+}_{0,3}$, so we have

$$(2) \qquad G \underset{\underline{\beta}_0}{\overset{\underline{\alpha}_0}{\rightrightarrows}} \mathcal{C}^{+}_{0,3} \ ,$$

characterized by

$$(2') \qquad \begin{cases} u(\rho) = \operatorname{int}(\alpha_0)\rho & \alpha_0 = \underline{\alpha}_0(u) \\ u(\sigma) = \operatorname{int}(\beta_0)\sigma & \beta_0 = \underline{\beta}_0(u). \end{cases}$$

We consider

$$(3) \qquad \begin{array}{l} G(0) = G \cap \widetilde{M}_{0,3}(0) \\ \cap \\ G[0] = G \cap \widetilde{M}_{0,3}[0] = \operatorname{Norm}_G(L^{\mathcal{C}}_0) \end{array}$$

so that

$$(4) \qquad G = G(0).\mathcal{C}^{+\wedge}_{0,3} \qquad \text{(semidirect product)}$$

$$(5) \qquad G[0] = G(0).L^{\mathcal{C}}_0 \qquad \text{id}$$

$$(6) \qquad G[0] \cap \mathcal{C}^{+\wedge}_{0,3} = L^{\mathcal{C}}_0.$$

We have set

$$(7) \qquad L^{\mathcal{C}}_0 = \text{closed subgroup of } \mathcal{C}^{+\wedge}_{0,3} \text{ generated by } \varepsilon_0$$

and we define likewise $L^{\mathcal{C}}_1, L^{\mathcal{C}}_{\infty}$. [p. 494]

Consider a profinite group †M, and a (continuous) surjective homomorphism [scan 496]

$$(8) \qquad \varphi : G \longrightarrow {}^{\dagger}M.$$

We set

$$(9) \quad \begin{cases} \varphi(\mathcal{C}_{0,3}^{+\wedge}) = \mathcal{C} & \text{normal subgroup of } M \\ \varphi(G(0)) = {}^{\dagger}M(0) \\ \varphi(G[0]) = {}^{\dagger}M[0]. \\ \varphi(L_0^{\mathcal{C}}) = L_0^{\dagger} \end{cases}$$

$\dagger$

Thus these are closed subgroups of M, satisfying

$$(10) \qquad M(0) \subset M[0] \subset \mathrm{Norm}_M(L_0)$$
$$(11) \qquad L_0^{\dagger} \subset \mathcal{C} \cap M[0].$$

By abuse of language, we still denote by $\rho, \sigma, \varepsilon_0, \varepsilon_1$ … the images of these elements in $\mathcal{C}$, which therefore satisfy there the relations known in $\mathcal{C}_{0,3}^{+\wedge}$, in particular

$$(12) \qquad \sigma^2 = \rho^3 = 1, \qquad \sigma\rho = \varepsilon_0, \quad \rho\sigma = \varepsilon_1, \qquad \varepsilon_1 = \sigma(\varepsilon_0) = \rho(\varepsilon_0).$$

The restrictions of the maps α_0, β_0 of (2,2') to $M_{\rho,\sigma}[0]$ will rather be denoted α, β, where α, β are in fact maps

$$(13) \quad \begin{array}{c} SG \overset{\text{dfn}}{=} G \\ \text{by } L_0^{\mathcal{C}},\ \text{inverse} \\ \text{image of } G \\ \text{in } S\widetilde{M}_{0,3} \end{array} \quad \rightrightarrows \ \mathcal{C}_{0,3}^{+\wedge}$$
$$= \{(u, f) \in G \times \mathcal{C}_{0,3}^{+\wedge} \mid fu \in G[0]$$

of which we look at the restrictions to the subgroup ($\simeq M_{\rho,\sigma}[0]$) of SG defined by the relation $f = 1$.

Composing $\underline{\alpha}, \underline{\beta}$:

$${}^{\dagger}G[0] \rightrightarrows \mathcal{C}_{0,3}^{+\wedge}$$

with $\varphi \mid \mathcal{C}_{0,3}^{+\wedge}$, we find maps, again denoted $\underline{\alpha}, \underline{\beta}$ (where $\underline{\alpha}^M, \underline{\beta}^M$ in case of possible confusion)

$$(14) \qquad G[0] \underset{\underline{\beta}}{\overset{\underline{\alpha}}{\rightrightarrows}} \mathcal{C}$$

thus satisfying

$$(15) \quad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \mathrm{int}(\beta)\sigma \end{cases} \quad \begin{array}{l} \text{for } u \in M[0] \text{ of the form } \varphi(u_0), \\ \text{and } \alpha = \underline{\alpha}(u_0),\ \beta = \underline{\beta}(u_0). \end{array}$$

[p. 495]

Uncertainty retained from cancelled passages.

140-3-s496-D001 A

140-3-s496-D002 A

140-3-s496-D003 A

140-3-s496-D004 A

140-3-s496-D005 $\mathcal{C}$
140-3-s496-D006 Thus
140-3-s496-D007 $\mathcal{C}$
140-3-s496-D009 $M[0]$

[scan 497]

A priori, α and β depend on the choice of u_0 from which u comes not only on u — but [p. 551]

$$(16) \qquad \begin{cases} \alpha\rho(\alpha)^{-1} = [u, \rho] \\ \beta\sigma(\beta)^{-1} = [u, \sigma] \end{cases}$$

do not depend on it. If we put

$$(17) \qquad T_\rho = \underline{\mathrm{Centr}}_M(\rho) \qquad T_\sigma = \underline{\mathrm{Centr}}_M(\sigma)$$

we thus see that, modulo T_ρ resp. T_σ, α, β depend only on u.

Hypothesis $\alpha = \alpha(u_0), \beta = \beta(u_0)$ depend only on $u = \varphi(u_0)$ (for $u_0 \in \dagger G[0]$).

We shall express this hypothesis in a slightly different way, by introducing

$$(18) \qquad J = \mathrm{Ker}(G[0] \longrightarrow M[0]),$$

we thus want that for $\dagger$

$$u_0' = \delta_0 u_0 \quad \text{with } u_0 \in G[0],\ \delta_0 \in J, \quad \text{we have}$$
$$\alpha(u_0') = \alpha(u_0), \qquad \beta(u_0') = \beta(u_0)$$

Write

$$\alpha_0 = \underline{\alpha}(u_0), \quad \beta_0 = \underline{\beta}(u_0), \quad \alpha_0' = \underline{\alpha}(u_0'), \quad \beta_0' = \underline{\beta}(u_0'), \quad \overline{\alpha}_0 = \underline{\alpha}(\delta_0), \quad \overline{\beta}_0 = \underline{\beta}(\delta_1|\delta_0),$$

in G, so that we shall have, by the cocycle formulas of G

$$\alpha_0' = \delta_0(\alpha_0)\overline{\alpha}_0, \qquad \beta_0' = \delta_0(\beta_0)\overline{\beta}_0,$$

whence, applying φ and noting that

$$\varphi(\delta_0(\alpha_0)) = \underbrace{\varphi(\delta_0)}_{1}(\varphi(\alpha_0)) = \varphi(\alpha_0) \overset{\text{dfn}}{=} \alpha \quad \text{and likewise}$$
$$\varphi(\delta_0(\beta_0)) = \varphi(\beta_0) \overset{\text{dfn}}{=} \beta$$

$$(19) \qquad \alpha' = \dagger\alpha\overline{\alpha}, \qquad \beta' = \beta\overline{\beta} \qquad \text{in } \mathcal{C}$$

where $\alpha, \beta, \alpha', \beta', \overline{\alpha}, \overline{\beta}$ are the images under φ of the quantities α_0 etc in $\mathcal{C}^{+\wedge}_{0,3}$. $\dagger$ Thus the hypothesis means this

$$(20) \qquad \forall\, \delta_0 \in J \overset{\text{dfn}}{=} \mathrm{Ker}(G[0] \longrightarrow M[0]), \quad \text{we have} \quad \alpha^{\Pi}(\delta_0), \beta^{\Pi}(\delta_0) \in \mathrm{Ker}\varphi$$

$\dagger$ When this hypothesis is satisfied, we thus have maps [p. 496]

Uncertainty retained from cancelled passages.

140-3-s497-D001 M
140-3-s497-D002 α
140-3-s497-D004 φ
140-3-s497-D005 with
140-3-s497-D006 when

[scan 498]

$$(21) \qquad M[0] \underset{\underline{\beta}}{\overset{\underline{\alpha}}{\rightrightarrows}} \mathcal{C}$$

factoring (14), and thus satisfying

$$(22) \qquad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \mathrm{int}(\beta)\sigma \end{cases} \qquad \text{for } u \in M[0], \quad \alpha = \underline{\alpha}(u), \quad \beta = \underline{\beta}(u),$$

which † paraphrase (2').

Moreover, let us put

$$(23) \qquad \zeta|\xi = \alpha\rho(\alpha)^{-1}, \qquad \eta = \beta\sigma(\beta)^{-1}$$

the relations (22) can be written thus

$$(24) \qquad u(\rho) = \zeta|\xi\rho, \qquad u(\sigma) = \eta\sigma$$

and we shall have

$$(25) \qquad \zeta|\xi = [u, \rho], \qquad \eta = [u, \sigma]$$

and moreover

$$(26) \qquad \zeta|\xi = \eta\varepsilon_1|{\ell_1}^{\nu} \qquad \text{where } \nu = \frac{\mu - 1}{2} \in \widehat{\mathbb{Z}} \quad (\mu \text{ the multiplier of } u).$$

which comes from the single relation in $\mathcal{C}^{+}_{0,3}$.

(We assume to fix ideas that the homomorphism

$$\dagger \qquad G \xrightarrow{\chi^G} \widehat{\mathbb{Z}}^* \qquad (\text{“multiplier”})$$

factors through

$$(27) \qquad G \longrightarrow M \xrightarrow{\chi^M} \widehat{\mathbb{Z}}^*,$$

so that we have a notion of multiplier ($\in \widehat{\mathbb{Z}}^*$) for the $u \in M$.)

Consider the natural homomorphism

$$(28) \qquad \begin{array}{ccc} M[0] & \longrightarrow & \mathrm{Aut}(\mathcal{C}) : \\ u & \longmapsto & \mathrm{int}(u)|\mathcal{C} \end{array}$$

we see from (22) even|or even (24) that it is known when we know [p. 497]

Uncertainty retained from cancelled passages.

140-3-s498-D002 generalize|generalizing

140-3-s498-D003 (27)

[scan 499]

the functions $\underline{\alpha}, \underline{\beta} : M[0] \longrightarrow \mathcal{C}$, and even when we know only $\zeta|\xi, \eta : M[0] \longrightarrow \mathcal{C}$. If moreover we put [p. 552]

$$(29) \qquad M'[0] = \mathrm{Im}(M[0] \longrightarrow \mathrm{Aut}(\mathcal{C}))$$

these latter functions $\zeta|\xi, \eta$ are already defined on $M'[0]$ — whereas it is not clear that the same holds for α, β, when $M[0]$ does not act|does not operate faithfully on $\mathcal{C}$. We thus have a map

for † ρ, σ generate topologically $\mathcal{C}$

$(\zeta|\xi,\eta)$ from a certain subgroup $M'[0] \subset \mathrm{Norm}_{\mathrm{Aut}(\mathcal{C})}(L_0)$ to the set $\mathcal{R}$ of solutions of the equation in $\zeta|\xi,\eta$ (26) which we write|shall write simply

$$(30) \qquad \eta^{-1}\zeta|\xi \in L_1$$

a relation which implies (26) (with the precise form of the exponent ν) if we assume (for example) $\mathcal{C}|M$ big enough that $\mathcal{C}^{+\wedge}_{0,3} \xrightarrow[2]{} \mathrm{SL}|\mathrm{PSL}(2,\widehat{\mathbb{Z}})$ factors through $\mathcal{C}$ — which we shall assume, to make life easier for ourselves.

† More precisely, let †

$$(32) \qquad \left\{ \begin{array}{c} \dagger = \Sigma_{\mathcal{C},\rho,\sigma} \subset \mathcal{C}\times\mathcal{C} \\ \| \\ \left\{ (\zeta|\xi,\eta) \;\middle|\; \begin{array}{ll} \text{a)} & \zeta|\xi\rho \text{ conjugate to } \rho \quad \dagger\text{in } \mathcal{C} \\ \text{b)} & \eta\sigma \quad \text{——————} \quad \sigma \quad \text{——————} \\ \text{c)} & \eta^{-1}\zeta|\xi \in L_1 \quad \text{i.e.} \quad \exists \nu' \in \widehat{\mathbb{Z}} \text{ such that } \zeta|\xi = \eta\varepsilon_1^{\nu'} \end{array} \right\} \end{array} \right\}$$

m7

Marginal note. m7

$$\underbrace{[\alpha,\rho] = \alpha\rho(\alpha)^{-1}}$$

$$\left\{ \begin{array}{l} \text{i.e. } \zeta|\xi \text{ of the form } [\alpha,\rho] \\ \text{resp. } \eta \text{ of the form } [\beta,\sigma] = \beta\sigma(\beta)^{-1} \end{array} \right.$$

†

[p. 498]

Uncertainty retained from cancelled passages.

140-3-s499-D001 [illeg. 1]
140-3-s499-D002 $\mathcal{C}$
140-3-s499-D003 H)|II)
140-3-s499-D004 $\Sigma|\mathcal{R}$
140-3-s499-D005 $\mathcal{C}$
140-3-s499-D006 every $\mathcal{C}$ fixing must such|such $\zeta|\xi$ $\zeta|\xi$ satisfy|satisfies and

[scan 500]

and let $u \in \mathrm{Aut}(\mathcal{C})$ be an automorphism of $\mathcal{C}$, and let us put

$$(34) \qquad \left\{ \begin{array}{ll} \zeta|\xi = [u,\rho] = u\rho u^{-1}\rho^{-1} = u(\rho)\rho^{-1} & \text{i.e. } u(\rho) = \zeta|\xi\rho \\ \eta = [u,\sigma] = u(\sigma)\sigma^{-1} & \text{i.e. } u(\sigma) = \eta\sigma \end{array} \right.$$

then to say that $\zeta|\xi\rho$ $(= u(\rho))$ is conjugate to ρ means that u fixes the conjugacy class of ρ, to say that $\eta\sigma$ $(= u(\sigma))$ is conjugate of|to σ means that u fixes the conjugacy class of σ. Finally the condition $\eta^{-1}\zeta|\xi \in L_1$ can also be written, since

cf. (24),(25)

$$\eta^{-1}\zeta|\xi = (\sigma u(\sigma^{-1}))(u(\rho)\rho^{-1}) = \sigma u(\sigma^{-1}\rho)\rho^{-1}$$

$$= \sigma\left(u(\underbrace{\sigma^{-1}\rho}_{\varepsilon_0})\dagger \underbrace{\rho^{-1}\sigma}_{\varepsilon_0^{-1}} \right)$$

†

N.B. $\sigma^{-1} = \sigma$

$$= \sigma(u(\varepsilon_0)\varepsilon_0^{-1})$$

in the form

$$\sigma(u(\varepsilon_0)\varepsilon_0^{-1}) \in L_1 \qquad \text{i.e.} \qquad u(\varepsilon_0)\varepsilon_0^{-1} \in L_0$$

(since $L_0 = \sigma^{-1}(L_1)$) or again (since $\varepsilon_0 \in L_0$)

$$u(\varepsilon_0) \in L_0, \quad \text{i.e. (since } \varepsilon_0 \text{ generates } L_0)$$

m10

Marginal note. m10 In this argument, we have used only the relation $\sigma^{-1}\rho = \varepsilon_0$ and the fact that ε_0 generates L_0, and not that $\sigma^2 = 1$ or $\rho^3 = 1$ for example.

$$u(L_0) = L_0$$

i.e. that u normalizes L_0.

Thus let

$$(35) \qquad \left\{ \begin{array}{c} \widetilde{B}'_0 \subset \mathrm{Aut}(\mathcal{C}) \\ \| \text{ dfn} \\ \left\{ u \in \mathrm{Aut}(\mathcal{C}) \;\middle|\; \begin{array}{l} u \text{ fixes conjugacy class of } \rho \text{ and that of } \sigma \\ u \text{ normalizes } L_0 \end{array} \right\} \end{array} \right.,$$

so that we have a map (injective because ρ, σ generate $\mathcal{C}$)

$$(36) \qquad \begin{array}{ccc} \widetilde{B}'_0 & \hookrightarrow & \Sigma \\ u & \longmapsto & ([u,\rho] = u(\rho)\rho^{-1},\ [u,\sigma] = u(\sigma)\sigma^{-1}). \end{array}$$

[p. 499]

Uncertainty retained from cancelled passages.

140-3-s500-D001 σ

140-3-s500-D002 whence

[scan 501]

[p. 553]

$$(37) \qquad \left\| \begin{array}{l} \text{Fundamental hypothesis on } \mathcal{C},\ \text{equipped with } \rho, \sigma,\ \text{whence } \varepsilon_0 = \sigma^{-1}\rho, \\ \varepsilon_1 = \rho\sigma^{-1} = \sigma(\varepsilon_0) = \rho(\varepsilon_0) : \\ \text{a) } \mathcal{C} \text{ is generated topologically by } \rho, \sigma \\ \text{b) The map (36) } \dagger \text{ (injective by a)) is} \\ \quad \text{bijective.} \end{array} \right.$$

$\dagger$

Condition b) thus means that for $(\alpha, \beta) \in \mathcal{C} \times \mathcal{C}$, satisfying a relation

$$\alpha\rho(\alpha)^{-1} = \beta\sigma(\beta)^{-1}\varepsilon_1^{\nu'}, \qquad \nu''|\nu' \text{ suitable},$$

there exists $u \in \mathrm{Aut}(\mathcal{C})$ such that we have

$$\begin{array}{lll} [u,\rho] = \alpha\rho(\alpha)^{-1} & \text{i.e. } u(\rho) = \mathrm{int}(\alpha)(\rho) & \text{i.e. } u(\rho)|u\rho) = \zeta|\xi\rho\dagger \\ [u,\sigma] = \beta\sigma(\beta)^{-1} & \text{i.e. } u(\sigma) = \mathrm{int}(\beta)\sigma & \text{i.e. } u(\sigma) = \eta\sigma \end{array}$$

(we shall thus necessarily have $u \in B'_0$ }, and u will be unique if we have a)).

Thus, on the set of the|of $R|\mathcal{R}_{\rho,\sigma}$ of the $(\zeta|\xi, \eta)$ satisfying a) b) c), we shall have a group structure, +|and deduced|described by transport of structure from that of $\widetilde{B}'_{\dagger 0}$

etc.

[p. 500]

Uncertainty retained from cancelled passages.
140-3-s501-D002 is|by
140-3-s501-D003 $\zeta|\xi$ is|u
140-3-s501-D005 int$(\zeta|\xi)\rho$
140-3-s501-D006 ρ

[scan 502]

II Let us return to the situation above, forgetting (at least for the moment) G, φ etc. The groups $M_{\rho,\sigma}$ etc
m13

Marginal note. m13 For a systematic rectification of the notations that follow|following cf. page 564, remark.

Let $\mathcal{C}$ be a group — we place ourselves, for what follows, either in the category of discrete groups, or in that of profinite groups, as we choose. We shall (to fix ideas) take that of profinite groups. We suppose given

$$(1) \qquad \rho, \sigma \in \mathcal{C}$$

m14

Marginal note. m14 [i.e. we have between $\varepsilon_0 = \ell'_0$, $\sigma = \ell'_1$ and $\rho^{-1} = \ell'_\infty$ with relation (2) $\ell'_\infty \ell'_1 \ell'_0 = 1$]

we put

$$(3) \qquad \begin{aligned} &\varepsilon_0 = \sigma^{-1}\rho, \qquad \varepsilon_1 = \rho\sigma^{-1} = \sigma(\varepsilon_0) = \rho(\varepsilon_0) \\ &L_0 = \text{closed subgroup generated by } \varepsilon_0 \\ &L_1 = \underline{\qquad\qquad\qquad\qquad\qquad\qquad} \ \varepsilon_1 \end{aligned} \qquad \text{, i.}$$

thus $L_1 = \rho(L_0) = \sigma(L_0)$.

We consider

$$\mathrm{Aut}(\mathcal{C}) \xrightarrow[\dagger]{\Phi=(\underline{\zeta|\xi},\underline{\eta})} \dagger\, \mathcal{C} \times \mathcal{C}$$

(4) defined by

$$(5) \qquad \begin{cases} \underline{\zeta|\xi}(u) = [u,\rho] = u\rho u^{-1}\rho^{-1} = u(\rho)\rho^{-1} & \text{thus } u(\rho) = \zeta|\xi\rho \quad (\zeta|\xi = \underline{\zeta|\xi}(u)) \\ \qquad\quad = u\rho(u)^{-1} & \\ \underline{\eta}(u) = [u,\sigma] = u\sigma u^{-1}\sigma^{-1} = u(\sigma)\sigma^{-1} & \text{thus } u(\sigma) = \eta\sigma \quad (\eta = \underline{\eta}(u)) \\ \qquad\quad = u\sigma(u)^{-1}. & \end{cases}$$

Thus for $u \in \mathrm{Aut}(\mathcal{C})$, and $\zeta|\xi = \underline{\zeta|\xi}(u)$, $\eta = \underline{\eta}(u)$, we have

$$(6)|(5) \qquad \begin{cases} u \text{ fixes the } \mathcal{C}\text{-conjugation class of } \rho & \iff \zeta|\xi\rho \text{ conjugate to } \rho \\ & \iff \zeta|\xi = \alpha\rho(\alpha)^{-1} \quad (\alpha \in \mathcal{C}) \\ u \text{ fixes the } \mathcal{C}\text{-conjugation class of } \sigma & \iff \eta\sigma \text{ conjugate to } \sigma \\ & \iff \eta = \beta\sigma(\beta)^{-1} \quad (\beta \in \mathcal{C}) \\ u \text{ normalizes } L_0 & \iff \eta^{-1}\zeta|\xi \in L_1 \iff \zeta|\xi = \eta\varepsilon_1^\nu \text{ with } \nu \in \widehat{\mathbb{Z}} \text{ suitable.} \end{cases}$$

[p. 501]

Uncertainty retained from cancelled passages.
140-3-s502-D002 φ
140-3-s502-D003 $\mathcal{C}$

Let|Let † [scan 503]

$$(7)\qquad \begin{matrix} B_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{C}) \\ \| \\ \{u \mid \text{satisfying the three conditions (6|5)}\} \end{matrix} \qquad \text{[p. 554]}$$

$$(8)\qquad \Sigma_{\rho,\sigma} \subset \mathcal{C}\times\mathcal{C} \qquad \| \qquad \left\{ (\zeta|\xi,\eta) \;\middle|\; \begin{array}{c} \zeta|\xi = \alpha\rho(\alpha)^{-1},\quad \eta = \beta\sigma(\beta)^{-1} \quad (\alpha,\beta\in\mathcal{C}) \\ \text{i.e. } \zeta|\xi\rho \text{ conjugate to } \rho, \qquad \text{i.e. } \eta\sigma \text{ conjugate to } \sigma \\ \eta^{-1}\zeta|\xi \in L_1 \quad \text{i.e. } \zeta|\xi = \eta\varepsilon_1^{\nu} \quad (\nu\in\widehat{\mathbb{Z}}) \end{array} \right\},$$

so that $B_{\rho,\sigma}$ is the inverse image of $\Sigma_{\rho,\sigma}$ under (4), and we have an induced map

$$B_{\rho,\sigma} \xrightarrow{\varphi=\varphi_{\rho,\sigma}} \Sigma_{\rho,\sigma}. \qquad (9)$$

We suppose

$$(10)\qquad \boxed{\begin{cases} \text{a)} & \rho,\sigma \text{ generate } \mathcal{C} \text{ topologically} \\ \text{b)} & \text{the map (9) (injective by a))} \\ & \text{is surjective (thus bijective).} \end{cases}}$$

m15

Marginal note. [m15] Careful, this condition is † satisfied for $\mathcal{C} = \mathrm{SL}(2,\widehat{\mathbb{Z}})'$! (it imposes † $\varepsilon = \varepsilon_3(\mu)$ on the group γ on writing $\zeta|\xi,\eta$…)

We then transport by φ the group structure of $B_{\rho,\sigma}$ to $\Sigma_{\rho,\sigma}$. We put †

$$(11)\qquad u_{\zeta|\xi,\eta} = \varphi^{-1}(\zeta|\xi,\eta) \in B_{\rho,\sigma} \qquad ((\zeta|\xi,\eta)\in\Sigma_{\rho,\sigma}),$$

so that † $u_{\zeta|\xi,\eta}$ is characterized by

$$(12)\qquad \begin{cases} u_{\zeta|\xi,\eta}(\rho) = \zeta|\xi\rho \\ u_{\zeta|\xi,\eta}(\sigma) = \eta\sigma \end{cases}$$

Hypothesis b) therefore means that, as soon as $(\zeta|\xi,\eta)\in\Sigma_{\rho,\sigma}$ i.e. $\zeta|\xi\rho$ conjugate to ρ, $\eta\sigma$ conjugate to σ and $\eta^{-1}\zeta|\xi\in L_1$, there exists [p. 502]

Uncertainty retained from cancelled passages.

140-3-s503-D001 $B_{\rho,\sigma}$

140-3-s503-D002 not

140-3-s503-D003 a condition of possibility

140-3-s503-D005 $\zeta|\xi$ $\zeta|\xi$

140-3-s503-D006 we shall have

an automorphism $u_{\zeta|\xi,\eta}$ of $\mathcal{C}$ satisfying (12) (and which will be unique by (10) a)). The group law on $\Sigma_{\rho,\sigma}$ is made explicit as follows: [scan 504]

$$(13)\quad (\zeta|\xi',\eta')(\zeta|\xi,\eta) = (\zeta|\xi'',\eta'') \quad \text{with} \quad \begin{cases} \zeta|\xi'' = u_{\zeta|\xi',\eta'}(\zeta|\xi)\zeta|\xi' \\ \eta'' = u_{\zeta|\xi',\eta'}(\eta)\eta'. \end{cases}$$

Let us consider the homomorphism †

$$\mathcal{C} \longrightarrow \mathrm{Aut}(\mathcal{C})$$

$$g \longmapsto \mathrm{int}(g)$$

and the inverse image of $B_{\rho,\sigma}$ is none other than the normalizer of L_0 in $\mathcal{C}$, i.e. N_0. We then want to construct a group $G_{\rho,\sigma}$, containing $\mathcal{C}$ as an invariant subgroup and into which † $B_{\rho,\sigma}$ maps, so that the tautological action of $B_{\rho,\sigma}$ on $\mathcal{C}$ is deduced from this homomorphism, and $G_{\rho,\sigma}$ is generated by $\mathcal{C}$ and the image of $B_{\rho,\sigma}$, denoted $B'_{\rho,\sigma}$, which will contain † N_0.

†If the center of $\mathcal{C}$ is trivial i.e.

$$\mathcal{C} \hookrightarrow \mathrm{Aut}(\mathcal{C})$$

($\mathcal{C}$ normal subgroup), †

$$\longrightarrow \dagger$$

$$\longmapsto$$

it suffices to take the subgroup $G_{\rho,\sigma}$ of $\mathrm{Aut}(\mathcal{C})$ generated by $\mathcal{C}$ and $B_{\rho,\sigma}$.

† But the hypothesis that the center of $\mathcal{C}$ be|is trivial is not a happy one — for example it is not satisfied if $\mathcal{C} = \mathrm{SL}(2, \widehat{\mathbb{Z}})/\pm 1$. It seems to me that the introduction of N_0 (over which in practice we have no control) [p. 503]

Uncertainty retained from cancelled passages.

140-3-s504-D002 Let us introduce

140-3-s504-D004 $B_{\rho,\sigma}$

140-3-s504-D005 thus

140-3-s504-D006 To|To

140-3-s504-D007 $\mathcal{C}$

140-3-s504-D009 $\mathcal{C}$

140-3-s504-D012 is|has commutative but

was not judicious either, and the condition $B'_{\rho,\sigma} \supset N_0$ — it suffices to have $B'_{\rho,\sigma} \supset L_0$. I shall therefore consider [scan 505] [p. 555]

$$(14) \quad \Big\{ S_0 G_{\rho,\sigma} \overset{\text{dfn}}{=} B_{\rho,\sigma} \cdot \mathcal{C} \qquad \text{the product } \tfrac{1}{2} \text{ direct, and}$$

$$L_0 \overset{i}{\hookrightarrow} \dagger\, S_0 G_{\rho,\sigma}$$

$$g \longmapsto \mathrm{int}(g^{\mathcal{C}}) \cdot g^{-1}$$

the image of this homomorphism is a normal subgroup, for a) it commutes with $\mathcal{C}$ (trivial) and b) it is invariant under $B_{\rho,\sigma}$, for we shall have for $u \in B_{\rho,\sigma}$

$$\mathrm{int}(u)(i(g)) = \mathrm{int}(u)(\mathrm{int}(g) \cdot g^{-1}) = \mathrm{int}(u(g)) \cdot \underbrace{\mathrm{int}(u)(g^{-1})}_{u(g^{-1}) = u(g)^{-1}} = i(u(g))$$

(i.e. i commutes with the actions of $B_{\rho,\sigma}$, operating on L_0 by restriction to L_0 of the tautological|total operation of $B_{\rho,\sigma}$ on $\mathcal{C}$, and on $\dagger S_0 G_{\rho,\sigma} \supset B_{\rho,\sigma}$ by inner automorphisms).

We can therefore consider

$$(15) \qquad G_{\rho,\sigma} \overset{\text{dfn}}{=} S_0 G_{\rho,\sigma} / i(L_0)$$

then the inclusion homomorphisms

$$B_{\rho,\sigma} \hookrightarrow S_0 G_{\rho,\sigma}$$

,

$$\mathcal{C} \longrightarrow S_0 G_{\rho,\sigma}$$

give, by composition with

$$S_0 G_{\rho,\sigma} \longrightarrow G_{\rho,\sigma}$$

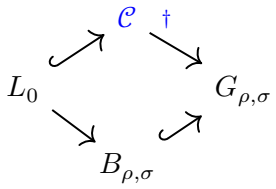
, homomorphisms

$$\dagger\mathcal{C} \xrightarrow{\dagger} G_{\rho,\sigma}$$

,

$$B_{\rho,\sigma} \hookrightarrow G_{\rho,\sigma}$$

(16) making the diagram commutative



(17) We note

$$\text{(18)} \qquad \mathrm{Ker}(\mathcal{C} \longrightarrow G_{\rho,\sigma}) = L_0 \cap \mathrm{Centr}(\mathcal{C}) = \mathrm{Ker}(L_0 \longrightarrow B_{\rho,\sigma})$$

[p. 504]

Uncertainty retained from cancelled passages.

140-3-s505-D002 $\mathcal{C}$
140-3-s505-D003 $\mathcal{C}$
140-3-s505-D004 $B_{\rho,\sigma}$
140-3-s505-D005 ‹
140-3-s505-D006 ‹

In the case $\mathcal{C} = \mathrm{SL}(2, \widehat{\mathbb{Z}})'$, (with ρ, σ as usual) we shall have $L_0 \cap \mathrm{Centr}(\mathcal{C}) = \{1\}$. Thus it does not seem a priori that there are any major drawbacks to supposing that [scan 506]

$$\text{(18)} \qquad L_0 \cap \mathrm{Centr}(\mathcal{C}) = \{1\} \quad \text{i.e.}$$

$$\mathcal{C} \hookrightarrow G_{\rho,\sigma} \quad \text{injective}$$

Let us note that $\dagger$ the image of

$$L_0 \longrightarrow B_{\rho,\sigma}$$

$$\ell \longmapsto \mathrm{int}(\ell)$$

(composite of (14) and the projection $S_0 G_{\rho,\sigma} \longrightarrow B_{\rho,\sigma}$) is obviously invariant|invariant, and putting

$$\Gamma^*_{\rho,\sigma}|\gamma_{\rho,\sigma} = B_{\rho,\sigma}/\operatorname{Im} L_0 \tag{19}$$

we find

$$G_{\rho,\sigma}/\mathcal{C}_{\rho,\sigma} \simeq B_{\rho,\sigma}/L_0 \simeq \Gamma|\gamma_{\rho,\sigma}. \tag{20}$$

m16

Marginal note. m16 i.e. homomorphism of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \dagger L_0 & \longrightarrow & B_{\rho,\sigma} & \to & \Gamma|\gamma_{\rho,\sigma} & - & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \mathcal{C} & \longrightarrow & G_{\rho,\sigma} & \to & \Gamma|\gamma_{\rho,\sigma} & - & 1 \\ & & \| & & & & & & \\ & & SG_{\rho,\sigma} & & & & & & \end{array}$$

(20bis)

(N.B. In the case $\mathcal{C} = \mathrm{SL}(2,\widehat{\mathbb{Z}})'$; † etc.|and, $B_{\rho,\sigma} \subset \mathrm{GP}(\widehat{\mathbb{Z}})$ is the group previously denoted B'_0 (or rather its isomorphic image $\widetilde{B}'_0$), and the homomorphism dét | $B'_0 = B_{\rho,\sigma}$ (trivial on L_0) induces by passage to the quotient an isomorphism

$$\Gamma|\gamma_{\rho,\sigma} \xrightarrow{\sim} G_{\mathrm{m}}$$

(21) m17

Marginal note. m17 In the case where we start from $\mathrm{SL}(2,\widehat{\mathbb{Z}})$, ρ, σ, we end up with $G = \mathrm{GL}(2,\widehat{\mathbb{Z}})$...

and we see, using the isomorphism

$$B'_0 = \tilde{T}_0|\tilde{\Gamma}_0 \cdot L_0 \quad (\tfrac{1}{2} \text{ direct})$$

where $\tilde{T}_0|\overline{T}_0 \xrightarrow{\sim} B'_0/L_0 \simeq G_{\mathrm{m}}$

i.e. $$B_{\rho,\sigma} \simeq \Gamma|\gamma_{\rho,\sigma} \cdot L_0,$$

that in this case we have [p. 505]

$$G_{\rho,\sigma} = \dagger \mathrm{GL}(2,\widehat{\mathbb{Z}})'. \tag{22}$$

[p. 505]

Uncertainty retained from cancelled passages.

140-3-s506-D001 is going to be|is its

140-3-s506-D002 $\mathcal{C}_{\rho,\sigma}$

140-3-s506-D003 $B_{\rho,\sigma}$

140-3-s506-D006 GL

[scan 507]

But in the more general cases to which we aspire, one must not at all expect $\Gamma_{\rho,\sigma}$ to be so small — for example that it be commutative. For example in the "universal" case, $\mathcal{C} = \mathcal{C}^{+\wedge}_{0,3}$, we will have [p. 556]

$$B_{\rho,\sigma} = \widetilde{M}_{0,3}[0]' = \operatorname{Ker}(\widetilde{M}_{0,3}[0] \xrightarrow{\varepsilon_3} \{\pm 1\}) \tag{23}$$

hence

$$\Gamma_{\rho,\sigma} \simeq \tilde{\Gamma}'_{0,3} \quad (\supset \Gamma'_{\mathbb{Q}}\ !). \tag{24}$$

On the other hand, we should be able without inconvenience to postulate the existence of a canonical splitting of the extension $B_{\rho,\sigma}$ of $\Gamma_{\rho,\sigma}$ by L_0. In fact, if we assume, † † in the general case, where nevertheless we assume $\sigma^2 = \rho^3 = 1$, i.e. $\mathcal{C}$ a quotient of $\mathcal{C}^{+\wedge}_{0,3}$, that the quotient $\mathcal{C}$ of $\mathcal{C}^{+\wedge}_{0,3}$ is large enough to map to GP|CP($\widehat{\mathbb{Z}}$), then † $B_{\rho,\sigma} \simeq \Sigma_{\rho,\sigma}$ maps to $\widetilde{B}'_{0,3}|\widetilde{B}'_0$, and the inverse image of $\tilde{T}_0$ is a subgroup (which we could denote $T_{\rho,\sigma}$)

$$T_{\rho,\sigma} \subset B_{\rho,\sigma} \qquad T_{\rho,\sigma} \xrightarrow{\sim} \Gamma_{\rho,\sigma} \tag{25}$$

which is a section subgroup, for $B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$, and also (as a subgroup of $G_{\rho,\sigma}$) for $G_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$, so that we will have

$$B_{\rho,\sigma} \simeq T_{\rho,\sigma}.L_0, \quad G_{\rho,\sigma} \simeq T_{\rho,\sigma}.\mathcal{C} \qquad (\tfrac{1}{2}\text{-direct products}) \tag{26}$$

[p. 506]

Uncertainty retained from cancelled passages.

140-3-s507-D001 $\mathcal{C}$ arbitrary

140-3-s507-D002 that [illeg. 1] is

140-3-s507-D003 that the

[scan 508]

Now let

$$\left\{ \begin{array}{c} S_0 g^{\dagger}_{\rho,\sigma} \dagger \subset \mathcal{C} \times \mathcal{C} \\ \| \text{ dfn} \\ \left\{ (\alpha,\beta) \;\middle|\; \underbrace{\alpha\rho(\alpha)^{-1}}_{\zeta|\xi} = \underbrace{\beta\sigma(\beta)^{-1}}_{\eta} \varepsilon_1^{\nu}, \text{ for } \nu \in \widehat{\mathbb{Z}} \text{ suitable} \right\} \\ \text{i.e.} \qquad \eta^{-1}\zeta|\xi \in L_1 \end{array} \right. \tag{27}$$

We have a canonical map

$$\left\{ \begin{array}{ccc} S_0 g_{\rho,\sigma} \dagger & \longrightarrow & \Sigma_{\rho,\sigma} \\ (\alpha,\beta) & \longmapsto & (\zeta|\xi,\eta), \quad \begin{array}{l} \zeta|\xi = \alpha\rho(\alpha)^{-1} \\ \eta = \beta\sigma(\beta)^{-1} \end{array}, \end{array} \right. \tag{28}$$

which is a surjection by definition of $\Sigma_{\rho,\sigma}$, and in fact by hypothesis (10) we have almost a canonical section

$$\begin{array}{ccc} \Sigma_{\rho,\sigma} & \longrightarrow & S_0 g_{\rho,\sigma} \dagger \\ \dagger & & \\ (\zeta|\xi,\eta) & \longmapsto & u_{\zeta|\xi,\eta} \end{array}$$

except that $u_{\zeta|\xi,\eta}$ is not in $\mathcal{C}$, but only in $G_{\rho,\sigma}$. We are going to consider the sets|the set of larger size

$$\begin{array}{c} g_{\rho,\sigma} \subset G_{\rho,\sigma} \times G_{\rho,\sigma} \\ \| \text{ dfn} \\ \left\{ (\alpha\beta) \;\middle|\; \underbrace{\alpha\rho(\alpha)^{-1}}_{\zeta|\xi} = \underbrace{\beta\sigma(\beta)^{-1}}_{\eta} \varepsilon_1^{\nu} \quad \text{for } \nu \in \widehat{\mathbb{Z}} \text{ suitable} \right\} \\ \dagger\zeta|\xi \end{array} \tag{29}$$

m18

Marginal note. m18 N.B. We will also have $\zeta|\xi = \alpha(\rho)\rho^{-1}$, $\eta = \beta(\sigma)\sigma^{-1}$ hence $\zeta|\xi, \eta \in \mathcal{C}$ (since $\mathcal{C}$ invariant in $G_{\rho,\sigma}$)

so that we will have

$$Sg_{\rho,\sigma} = g_{\rho,\sigma} \cap (\mathcal{C} \times \mathcal{C}). \tag{30}$$

(assuming (18) to simplify). [p. 507]

Uncertainty retained from cancelled passages.

140-3-s508-D001 $\wedge$
140-3-s508-D002 $\mathcal{C}$
140-3-s508-D003 $\mathcal{C}$
140-3-s508-D005 (29)
140-3-s508-D006 $\mathcal{C}$
140-3-s508-D007 η^{-1}

[scan 509]
[p. 508]
[p. 557]

We find again

$$\begin{cases} g_{\rho,\sigma} \longrightarrow \Sigma_{\rho,\sigma} \\ (\alpha,\beta) \longmapsto (\zeta|\xi, \eta), \quad \begin{matrix} \zeta|\xi & = \alpha\rho(\alpha)^{-1} = [\alpha,\rho] = \alpha(\rho)\rho^{+1|-1} \\ \eta & = \beta\sigma(\beta)^{-1} = [\beta,\sigma] = \beta(\sigma)\sigma^{-1} \end{matrix} \end{cases} \tag{31}$$

(by definition of $B_{\rho,\sigma}$ we do indeed find that the $(\zeta|\xi, \eta)$ associated with the (α, β) are in $\Sigma_{\rho,\sigma}$). This time hypothesis (10) implies that we have a canonical section

$$\begin{matrix} \Sigma_{\rho,\sigma} \longrightarrow g_{\rho,\sigma} \\ (\zeta|\xi, \eta) \longrightarrow (u_{\zeta|\xi,\eta}, u_{\zeta|\xi,\eta}) \end{matrix} \tag{32}$$

and we find the same|the statements of § 48 III).

Theorem Let us consider subgroups of $\dagger G_{\rho,\sigma}$

$$\dagger \quad \begin{cases} Z_\rho = \underline{\mathrm{Centr}}_{G_{\rho,\sigma}}(\rho) \\ Z_\sigma = \underline{\mathrm{Centr}}_{G_{\rho,\sigma}}(\sigma) \end{cases}, \tag{33}$$

m19

Marginal note. m19 (N.B. in the case $\mathcal{C} = \mathrm{SL}(2, \widehat{\mathbb{Z}})'$ etc., we will have

$$Z_\rho = T_\rho, \qquad Z_\sigma = N'_\sigma$$

and the product group

$$B_{\rho,\sigma} \times T_\rho \times T_\sigma \dagger$$

then we have a $\dagger$ bijective map

$$\begin{matrix} B_{\rho,\sigma} \times T_\rho|Z_\rho \times T_\sigma|Z_\sigma \dagger \xrightarrow{\sim} g_{\rho,\sigma} \\ (u,a,b) \longmapsto (ua^{-1}, ub^{-1}) \end{matrix} \tag{34}$$

m20

Marginal note. m20 by definition of $B_{\rho,\sigma}$ (7) and we see that $Z_\rho \to B_{\rho,\sigma}$ and $Z_\sigma \to B_{\rho,\sigma}$ are epimorphisms, hence also their composites $\delta_\rho, \delta_\sigma$ with $B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$. hence the maps (35) are epimorphisms.

Corollary. Let us consider the group homomorphisms $\delta_B : B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$,

$$\{\delta_G : G_{\rho,\sigma} \longrightarrow \Gamma_{\rho,\sigma}, \quad \delta_\rho : Z_\rho \longrightarrow \Gamma_{\rho,\sigma}, \quad \delta_\sigma : Z_\sigma \longrightarrow \Gamma_{\rho,\sigma}, \tag{35}$$

$\delta_B : B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$ δ_G is the canonical homomorphism of passage to the quotient $\mathcal{C}_{\rho,\sigma} = \mathcal{C}$, and $\delta_\rho, \delta_\sigma, \delta_B$ are the composites of δ_G with the inclusions $Z_\rho \subset G_{\rho,\sigma}$, $Z_\sigma \hookrightarrow G_{\rho,\sigma}$, $B_{\rho,\sigma} \subset G_{\rho,\sigma}$

Uncertainty retained from cancelled passages.

140-3-s509-D003 g

140-3-s509-D004 $Z_{\rho,\sigma}$

140-3-s509-D005 $= S\widetilde{\Gamma}_{\rho,\sigma}$

140-3-s509-D006 the

140-3-s509-D007 $S\widetilde{\Gamma}_{\rho,\sigma}$

m21

[scan 510]

Marginal note. m21 denoted $S_0\Gamma^b_{\rho,\sigma}$ in the § 48, IX ff

Let

$$\begin{gathered} \dagger S_0\Gamma^{b_0}_{\rho,\sigma} \subset B_{\rho,\sigma} \times Z_\rho \times Z_\sigma \\ \| \text{ dfn} \\ \{(u,a,b) \mid \delta_G(u) = \delta_\rho(a) = \delta_\sigma(b)\}. \end{gathered} \tag{36}$$

† i.e.

$$S\Gamma'_{\rho,\sigma} = B_{\rho,\sigma} \times_{\Gamma_{\rho,\sigma}} Z_\rho \times_{\Gamma_{\rho,\sigma}} Z_\sigma. \tag{37}$$

a bijection †

$$\begin{gathered} S_0\Gamma^{b_0}_{\rho,\sigma} \xrightarrow{\sim} S_0 g_{\rho,\sigma} \quad \text{(defined in (27))} \\ (u,a,b) \longmapsto \dagger(\zeta|\xi,\eta), \quad \text{with } \zeta|\xi = ua^{-1},\ \eta = ub^{-1}. \end{gathered} \tag{38}$$

m22

Marginal note. m22 N.B. The letter S here is inspired by the usage of Mister Ahlfors for the "special Teichmüller groups", and does not at all have the same meaning as in SZ_ρ, SZ_σ, where it indicates that we take the kernels of $\delta_\rho, \delta_\sigma$ (playing the role of a "determinant"…)

⟹

Scholium By the bijections (34) (38), we transport onto $g_{\rho,\sigma}$, $S_0 g_{\rho,\sigma}$ the natural group structures of $\tilde{\Gamma}_{\rho,\sigma}$, $\Gamma^{b_0}_{\rho,\sigma}$. Making this group structure explicit we find in particular for $Sg_{\rho,\sigma}$

$$\begin{cases} (\alpha',\beta')(\alpha,\beta) = \alpha''\beta'', \quad \text{with } \alpha'' = u'(\alpha)\alpha', \quad \beta'' = u'(\beta)\beta'. \\ u' = u_{\zeta|\xi',\eta'} : \quad \mathcal{C} \longrightarrow \mathcal{C}\ , \quad \text{defined by } \begin{cases} u(\rho) = \zeta|\xi\rho = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \eta\sigma = \mathrm{int}(\beta)\sigma \end{cases} \end{cases} \tag{39|37}$$

(with $\zeta|\xi = \alpha\rho(\alpha)^{-1}$, $\eta = \beta\sigma(\beta)^{-1}$).

In $S_0\Gamma^{b_0}_{\rho,\sigma}$ we have a subgroup isomorphic to L_0, i.e. m23

Marginal note. m23

$$\begin{aligned} L_0 &\longrightarrow S\Gamma'_{\rho,\sigma} \\ \ell &\longmapsto (\ell, 1, 1) \end{aligned} \tag{39}$$

and we have exact sequence

$$1 \to L_0 \to S\Gamma'_{\rho,\sigma} \to \Gamma_{\rho,\sigma} \to 1 \tag{40}$$

†

[p. 509]

Uncertainty retained from cancelled passages.

140-3-s510-D001 $S\widetilde{\Gamma}^0_{\rho,\sigma} \quad = \widetilde{M}^0_{\rho,\sigma} \quad \subset S_0\Gamma_{\rho,\sigma}$

140-3-s510-D002 $S\Gamma'_{\rho,\sigma}$

140-3-s510-D003 here

140-3-s510-D004 (ua^{-1}, ub^{-1})

140-3-s510-D005 Let us now put $\mathcal{C}$ a)|1°) *which* $\mathcal{C}$ $\mathcal{C}$ such that b)|2°) $\mathcal{C}$ $\mathcal{C}$ $\mathcal{T}|\Gamma$ $\mathcal{T}|\Gamma$ $\delta_{\mathcal{T}}$